9. Let A and B be two sets such that $n(A) = 3$ and $n(B) = 2$. If $(x, 1), (y, 2), (z, 1)$ are in $A \times B$, find A and B, where x, y, z are distinct elements.
10. If $A \times A$ has 16 elements, $S = \{(a, b) \in A \times A : a < b\}$; $(-1, 2)$ and $(0, 1)$ are two elements of S, then find the remaining elements of S.

1.4 Constants and Variables, Intervals and Neighbourhoods

To continue our discussion, we need certain prerequisites namely, constants, variables, independent variables, dependent variables, intervals and neighbourhoods.

1.4.1 Constants and Variables

A quantity that remains unaltered throughout a mathematical process is called a ***constant***. A quantity that varies in a mathematical process is called a ***variable***. A variable is an ***independent variable*** when it takes any arbitrary (independent) value not depending on any other variables, whereas if its value depends on other variables, then it is called a ***dependent variable***.

We know the area A of a triangle is given by $A = \frac{1}{2}bh$. Here $\frac{1}{2}$ is a constant and A, b, h are variables. Moreover b and h are independent variables and A is a dependent variable. We ought to note that the terms *dependent* and *independent* are relative terms. For example in the equation $x + y = 1$, x, y are variables and 1 is a constant. Which of x and y is dependent and which one is independent? If we consider x as an independent variable, then y becomes dependent whereas if we consider y as an independent variable, then x becomes dependent.

Further consider the following examples:

(i) area of a rectangle $A = \ell b$.
(ii) area of a circle $A = \pi r^2$.
(iii) volume of a cuboid $V = \ell bh$.

From the above examples we can directly infer that b, h, ℓ, r are independent variables; A and V are dependent variables and π is a constant.

1.4.2 Intervals and Neighbourhoods

The system $\mathbb{R}$ of real numbers can be represented by the points on a line and a point on the line can be related to a unique real number as in Figure 1.2. By this, we mean that any real number can be identified as a point on the line. With this identification we call the line as the ***real line***.



Figure 1.2

The value increases as we go right and decreases as we go left. If x lies to the left of y on the real line then $x < y$. As there is no gap in a line, we have infinitely many real numbers between any two real numbers.

Definition 1.1

A subset I of $\mathbb{R}$ is said to be an ***interval*** if

(i) I contains at least two elements and
(ii) $a, b \in I$ and $a < c < b$ then $c \in I$.

Geometrically, intervals correspond to rays and line segments on the real line.